

Extreme Climate Spells and Urban Resilience in Bogotá, Colombia

A Spatially Explicit Climate–Impact Analysis

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PhD Research Proposal – Draft

Background

Recent patterns in anthropogenic activities are altering the variability, frequency, persistence, and intensity of extremes [1, 2, 3]. Temperature and precipitation extremes are especially relevant because their impacts can accumulate when anomalous conditions persist for several consecutive days [6, 7]. In Colombia, hydroclimatic variability is also strongly affected by large-scale processes such as ENSO, which influences precipitation and hydrological conditions across the Andes [5, 4].

However, climatic hazard alone does not explain impact. Similar climatic events can produce very different consequences depending on population characteristics, urban form, land cover, infrastructure, socioeconomic conditions, and adaptive capacity [1, 8]. Bogotá is therefore an interesting case for studying how extreme climate conditions translate into spatially uneven human impacts.

Previous Latin American research has shown nonlinear complex relationships between temperature and mortality large differences among cities [10]. Bogotá also has official risk, environmental, mortality, and spatial datasets that could support a local analysis [11, 12, 13].

Main Question

When does a climate extreme becomes and impact and why do similar climatic events produce different impacts across space?

The project would focus on persistent temperature and precipitation anomalies, referred to here as *climate spells*, and relate them to human impacts and urban vulnerability in Bogotá.

Objectives

1. Identify changes in the frequency, intensity, and duration of extreme temperature and precipitation spells.
2. Relate climate spells to a measurable impact, with mortality as the main initial case.
3. Test whether the climate impact relationship varies spatially across Bogotá.
4. Explore whether socioeconomic, demographic, environmental, and urban variables help explain these differences.
5. Develop a spatially explicit resilience framework and test whether it can be extended to rainfall-related flooding.

Proposed Data

Climate information would initially come from ERA5-Land, potentially combined with IDEAM local meteorological observations [9]. Mortality data could be obtained from DANE Vital Statistics [11]. Additional variables could include age structure, population density, socioeconomic indicators, green space, impervious surface, access

to health services, and district risk information.

The final spatial unit would depend on data availability and confidentiality. Analysis could be performed at locality, UPZ, census-sector, or grid-cell level.

Methods

Extreme events would be defined using local percentile thresholds, for example temperatures above the 95th or 99th percentile or high precipitation percentiles. A spell would represent consecutive or near-consecutive days above the selected threshold. For each spell, duration, intensity, recurrence, timing, and cumulative magnitude could be calculated.

The first analytical stage would use relatively standard approaches such as correlation, generalized additive, distributed-lag, spatial regression, or hierarchical models. These would provide a baseline before testing more complex AI approaches [10].

Optional AI directions

One part I would particularly like to explore during the PhD is whether explicit spatial structure can improve the analysis. Possible methods include:

- **Geospatial knowledge graphs:** representing areas, climate events, deaths, infrastructure, and vulnerability indicators as connected entities [14, 15].
- **Graph or spatially explicit neural networks:** learning from both local characteristics and spatial/semantic relationships between areas [16, 17].
- **Clustering or representation learning:** identifying recurring climate-risk or resilience profiles without defining them in advance [18].
- **Fuzzy logic:** combining uncertain concepts such as high exposure, low adaptive capacity, or high vulnerability into interpretable resilience indicators [19, 20].

These methods are not intended as fixed commitments at this stage. A central methodological question would be whether adding explicit spatial and semantic structure improves prediction, interpretability, or understanding compared with simpler statistical models.

Expected Contribution

The project would move from simply detecting climatic extremes toward understanding how they interact with the urban system. The main output would be a spatial framework identifying areas where similar climate stress produces disproportionately high or low impacts. This could contribute to climate adaptation, public-health planning, risk management, and more targeted resilience strategies in Bogotá.

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